

Cost-Effectiveness Analysis

METIMMOX-1 Economic Evaluation

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1 Overview

This cost-effectiveness analysis evaluates standard of care against two pre-treatment biomarker-guided immunotherapy strategies for metastatic MSS/pMMR colorectal cancer:

- CRP-guided treatment selection
- TMB/BRAF-guided treatment selection

The economic survival model is a single joint model with CRP and TMB/BRAF treatment interactions. Economic strategies are restricted to pre-treatment biomarkers.

2 Economic Survival Model

Table 1: Single economic survival model

Component	Formula
OS	$\text{Surv}(\text{OSwk}, \text{Death}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} * \text{Rx} + \text{tmb_braf} * \text{Rx}$
PFS	$\text{Surv}(\text{PFSwk}, \text{Progression}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} * \text{Rx} + \text{tmb_braf} * \text{Rx}$
Control	$\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex}$

Table 2: Economic strategies

Strategy	Definition
Standard of Care	All patients receive standard FLOX chemotherapy
CRP-guided	CRP-positive patients receive FLOX + nivolumab; CRP-negative patients receive FLOX
TMB/BRAF-guided	TMB/BRAF-positive patients receive FLOX + nivolumab; TMB/BRAF-negative patients receive FLOX

3 Base Case Results

Table 3: Base case results at WTP = EUR 51,000

Strategy	Cost	QALYs	NMB
Standard of Care	EUR 21,056	1.350	EUR 47,783
CRP-guided	EUR 53,791	1.389	EUR 17,026
TMB/BRAF-guided	EUR 63,235	1.351	EUR 5,684

Table 4: Incremental cost-effectiveness results

Strategy	Cost	QALYs	Incremental Cost	Incremental QALYs	ICER	Stat
Standard of Care	EUR 21,056	1.350	–	–	–	ND
CRP-guided	EUR 53,791	1.389	EUR 32,735	0.039	EUR 843,973	ND
TMB/BRAF-guided	EUR 63,235	1.351	–	–	Dominated	D

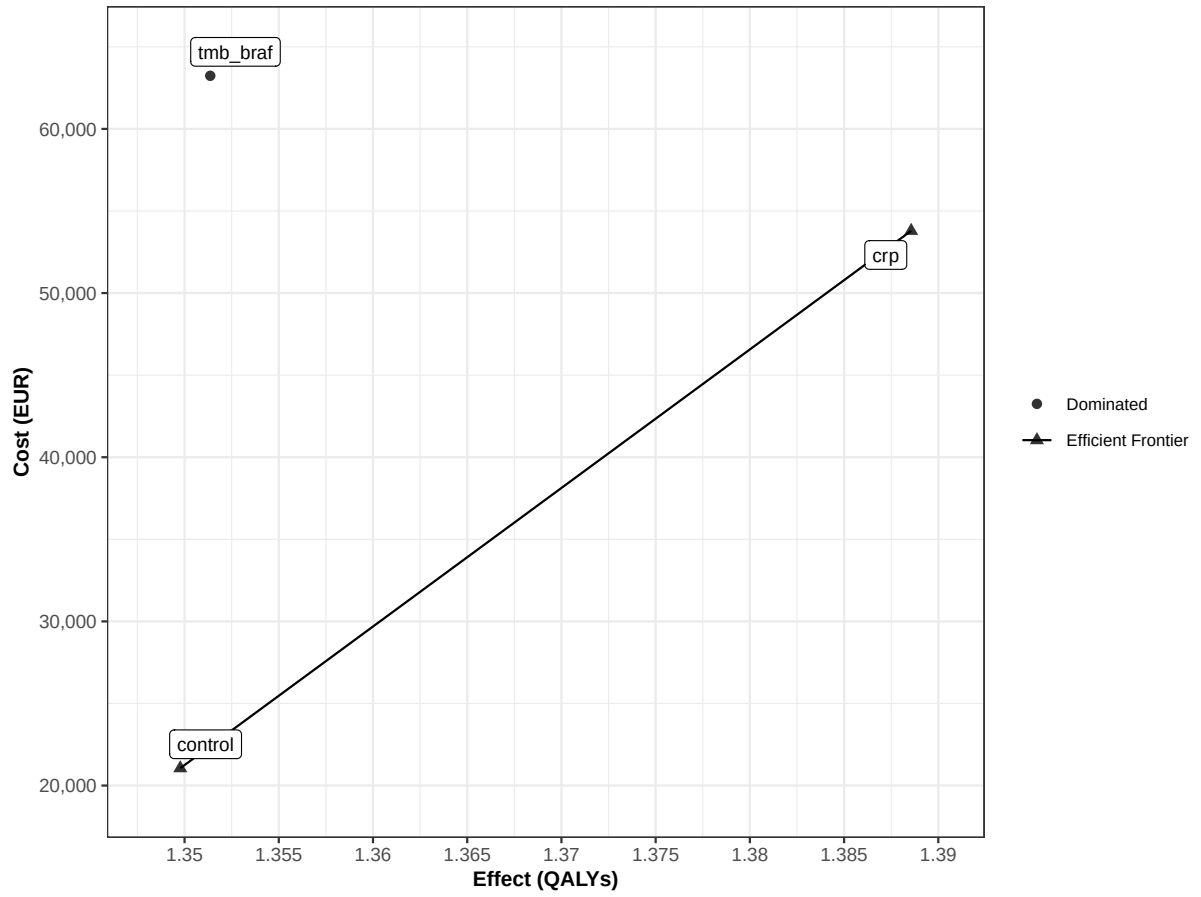


Figure 1: Cost-effectiveness plane for the single joint economic model

4 Probabilistic Sensitivity Analysis

Table 5: PSA summary at WTP = EUR 51,000

Strategy	Mean Cost	Mean QALYs	Probability Cost-Effective
Standard of Care	EUR 21,187	1.298	95.7%
CRP-guided	EUR 54,085	1.418	2.6%
TMB/BRAF-guided	EUR 62,615	1.395	1.8%

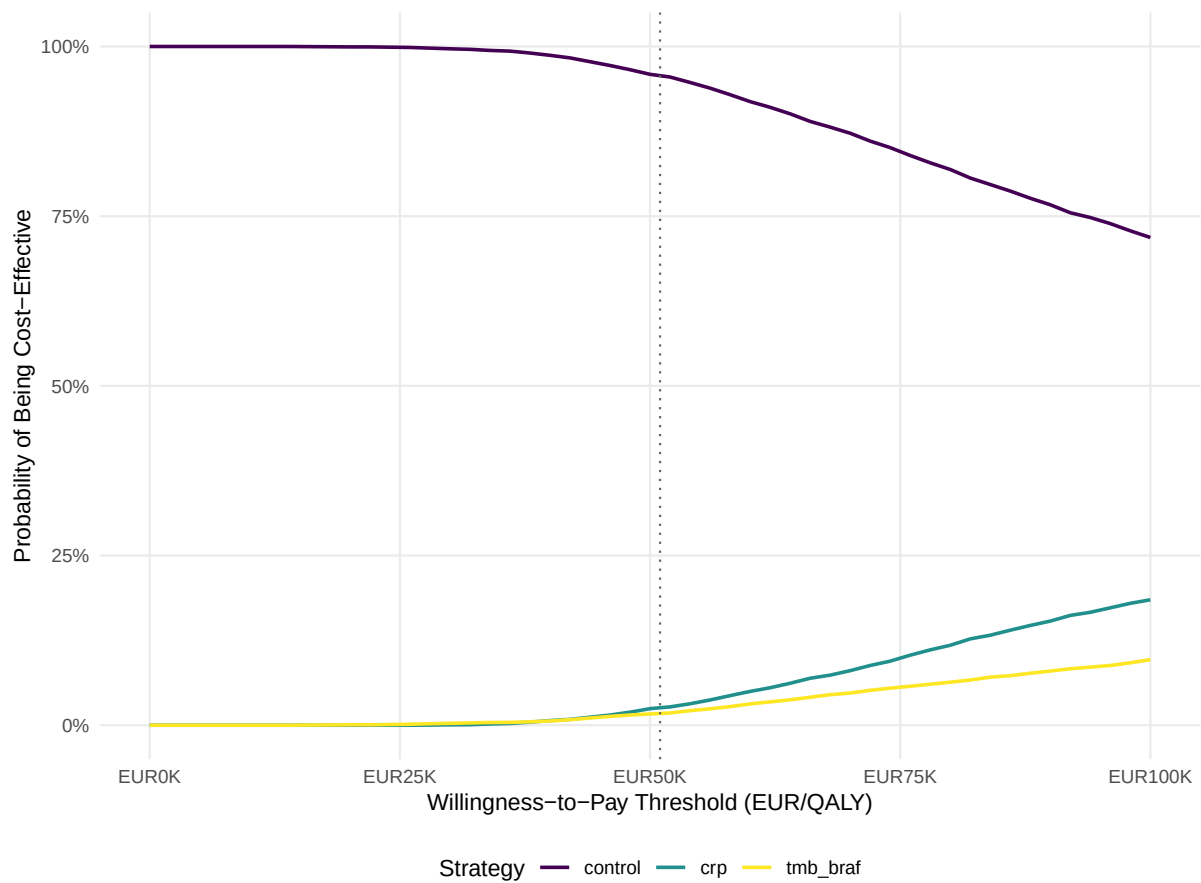


Figure 2: Cost-effectiveness acceptability curves

5 Summary

The economic evaluation now uses one joint survival model and three economic strategies: standard of care, CRP-guided immunotherapy, and TMB/BRAF-guided immunotherapy.

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